Timothy Alder

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GitHub: https://github.com/TjAlderman/tplat
Mobile: 0478 293 399
DOB: 16/12/2002

Bachelor of Engineering (R&D) Hons.

Electrical and Communication Systems Major with Mechatronics Minor

Seeking the opportunity to work with an employer who provides a clear and thoughtful career pathway. I am passionate about signal processing, computer vision, home automation, and martial arts.

RELEVANT TRAITS

- Multi-modal projects
- Scientific experimentation
- Time management skills
- Self-guided learning
- Research-oriented work
- Workflow automation
- Confidence in dealing with a variety of people
- Ongoing commitment to skills development

Software Skills: Python, MATLAB, Simulink, C, Verilog, Bazel, GitHub, Linux.

Libraries: OpenCV, Tensorflow, PyTorch, NumPy, Pandas, SciPy, scikit-learn, Qt.

Hardware Skills: oscilloscope; power supply; waveform generator; digital multimeter; optical posts, stages and mounts; spectrometer.

RELEVANT WORK EXPERIENCE

Undergraduate Optical Engineer (*Seeing Machines Ltd, Fyshwick*): May 2022 – present
Responsibilities included:

- Design of novel testing procedures to quantify and compare the performance of optical components.
- Programmatic automation, standardisation, and documentation of existing test procedures. Compiled and migrated existing fragmented test procedure documentation and scripts to a unified codebase distributable as a wheel file. Documentation hosted on Github pages.
- Novel test procedure automation workflows using Python. Combined multiple camera and hardware APIs to entirely automate camera calibration and image quality testing procedures using a 2-axis goniometer, linear rail, optical power meter, and a spectrometer. Included wrapping of some C APIs with pybind11 and ctypes.
- Design of testing procedures and subsequent analysis quantifying the relationship between algorithm performance and optical system metrics.
- Design and oversight of a large-scale data collection study involving >30 participants and >10 hours of video data.
- A novel 12-unit research project on neural-based skin classification using the contrast of subsurface scattering from coherent illumination sources.
- A novel 12-unit research project on neural-based remote photoplethysmography using structured near-infrared illumination.

Optical Engineering Team Internship (*Seeing Machines Ltd, Fyshwick*): Feb. 2022 – May 2022
Responsibilities included:

- Training on the use of optical laboratory equipment.
- Testing and evaluation of optical components and pathways.

EDUCATION AND QUALIFICATIONS

Australian National University – 2021-2024.

- 6.516 GPA
- First Class Honours
- University Medal
- Chancellor's Letter of Commendation
- Completed 3 individual research projects totalling 26 units.
- 3-month internship in the Optical Engineering Team at Seeing Machines.

- 2+ years working as an Undergraduate Optical Engineer at Seeing Machines.

Marist College Canberra – graduated 2020.

- National Youth Science Forum alumni.
- 100 Hours Community Service Award.
- College House Vice Captaincy.
- ANU Australian Excellence Scholarship recipient.
- 98.55 ATAR (Proxime Accessit).

REFEREES

Mr. Charles Crespin	**Mr. Scott McKenzie**	**Dr. Andrei Komar**
Optics Team Lead	Lieutenant Colonel	Senior Optical Scientist
(Seeing Machines Ltd.)	*(Australian Army)*	(*Seeing Machines Ltd.*)
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